

September 17, 2021

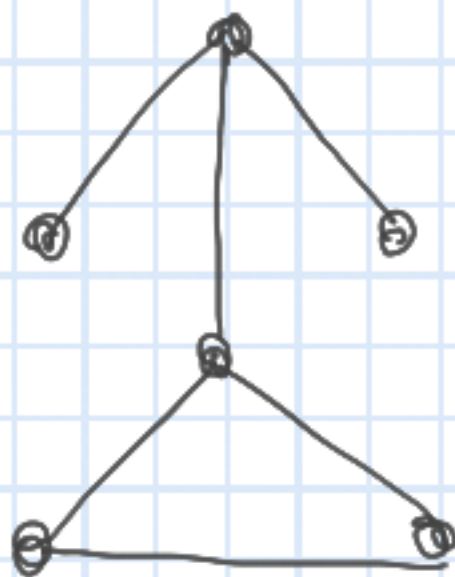
Graph algorithms

- ① Brief review of graphs
- ② How to store a graph in computer
memory
- ③ Traversals
- ④ Minimum spanning trees

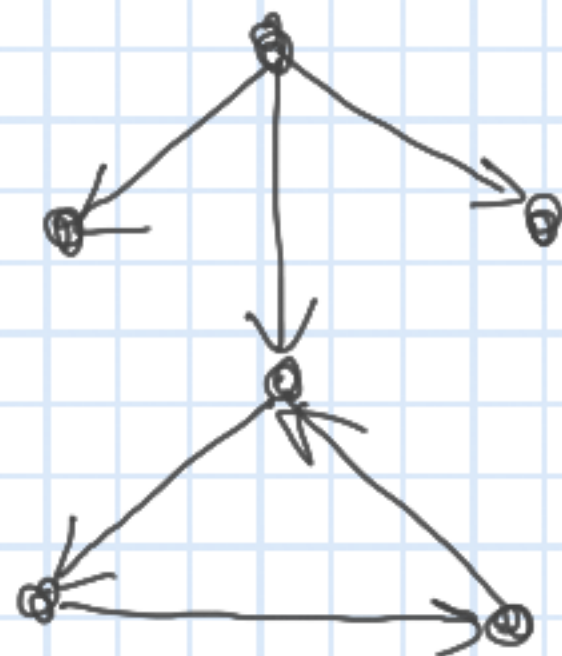
⑤ Shortest path (GPS)

Review

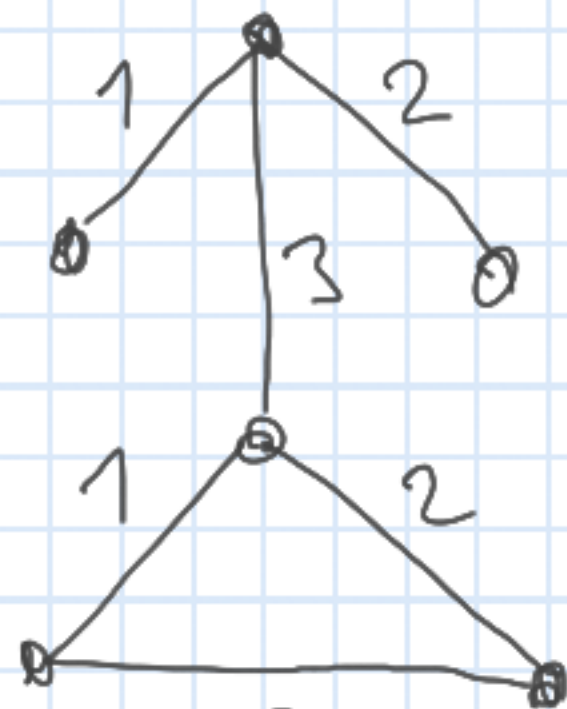
Graph



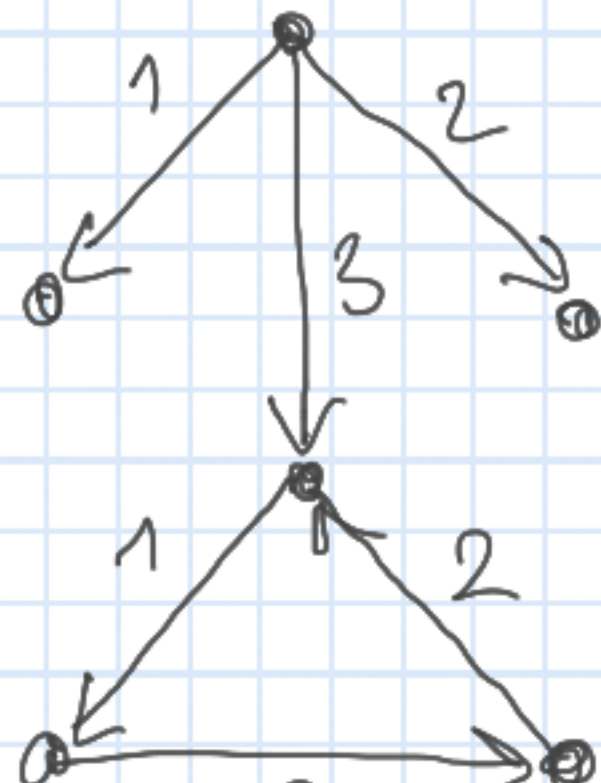
undirected
graph



directed
graph



weighted
undirected g.



weighted
directed g.

Mathematically of DiMat

Graph : $V \neq \emptyset \rightarrow$ vertices

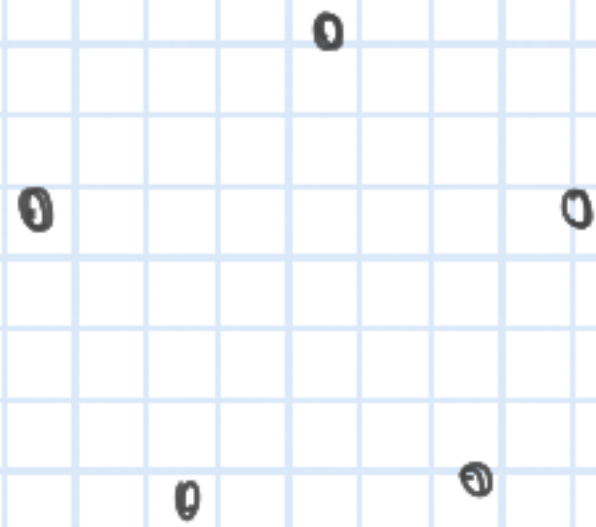
$E \rightarrow$ edges

adjacency relation

$w : E \rightarrow \mathbb{R} \rightarrow$ weight function

Some special graphs
(cf. DiMat)

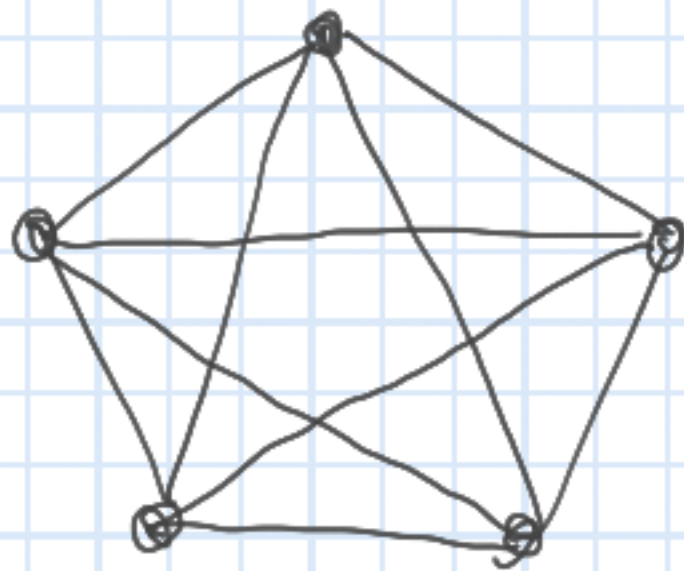
(A)



$$E = \emptyset$$

empty graph

(B)



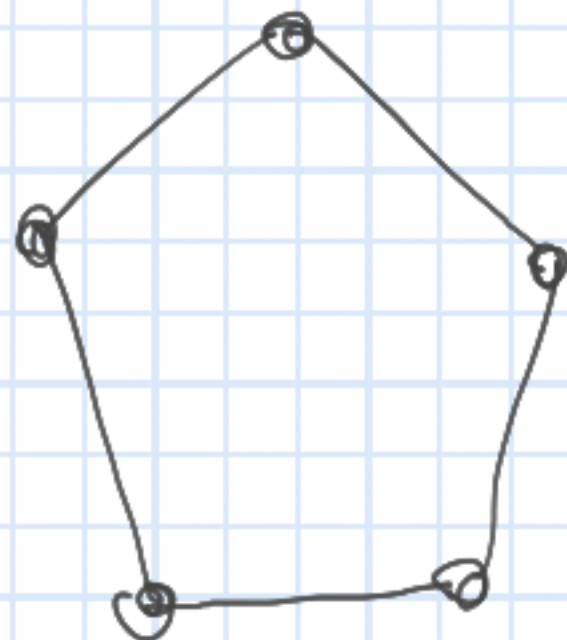
every two vertices are
connected by an edge
complete graph

in a complete graph $|E| = \frac{|V|(|V|-1)}{2} = \Theta(|V|^2)$

Remark

- graphs in which $|E|$ is close to $\frac{|V|(|V|-1)}{2}$ are called dense graphs
- graphs in which $|E| = O(|V|)$ are called sparse graphs

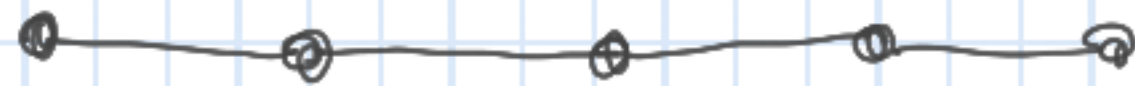
(C)



cycle

(length = number of edges)

(D)



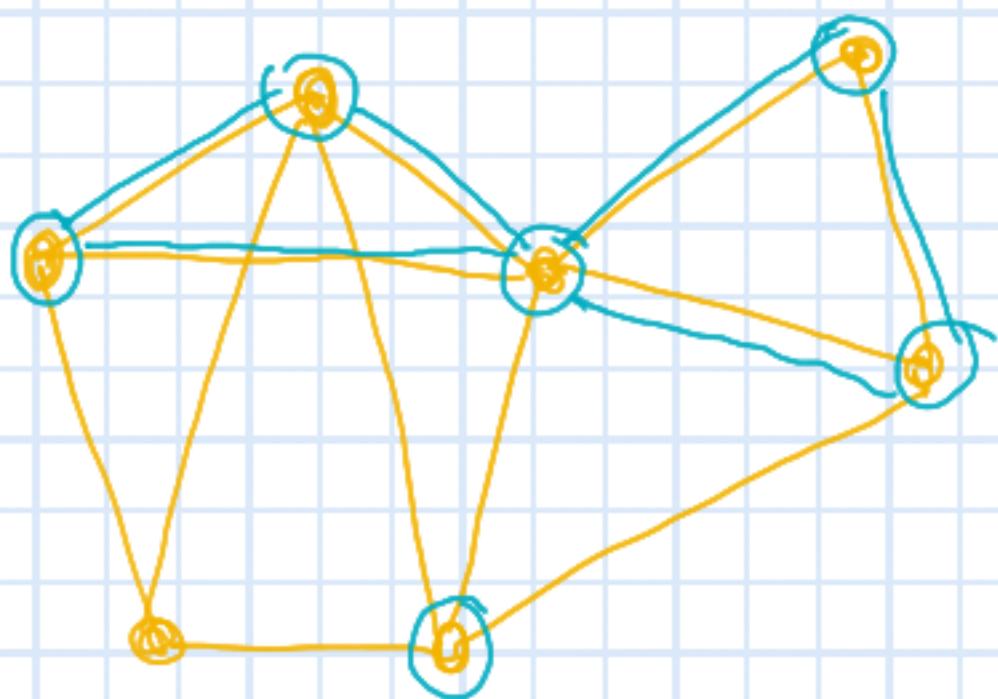
path

(length = number of edges)

Subgraph

Graph $G' = (V', E')$ is a subgraph of the

graph $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$



$$G = (V, E)$$

$$G' = (V', E')$$

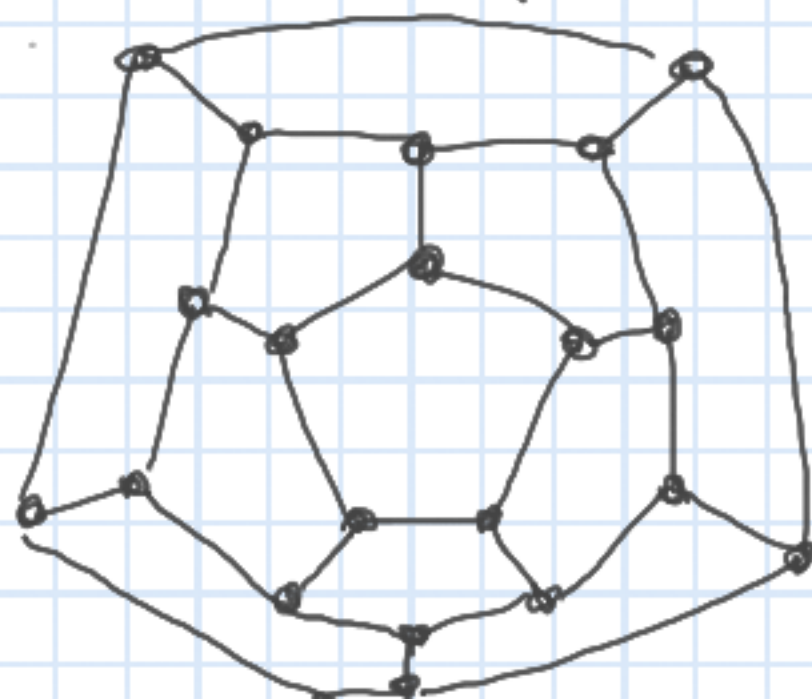
Challenge (\$1.000.000)

① Does there exist a cycle in a graph going through all vertices?

② Does there exist a path in a graph going through all vertices?

Hamiltonian cycle, Hamiltonian path

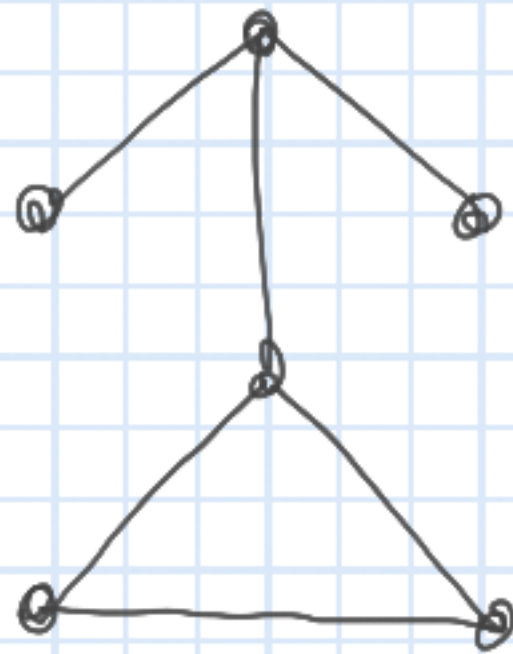
Dodecahedral game: Find a Hamiltonian cycle in the following graph (if exists)



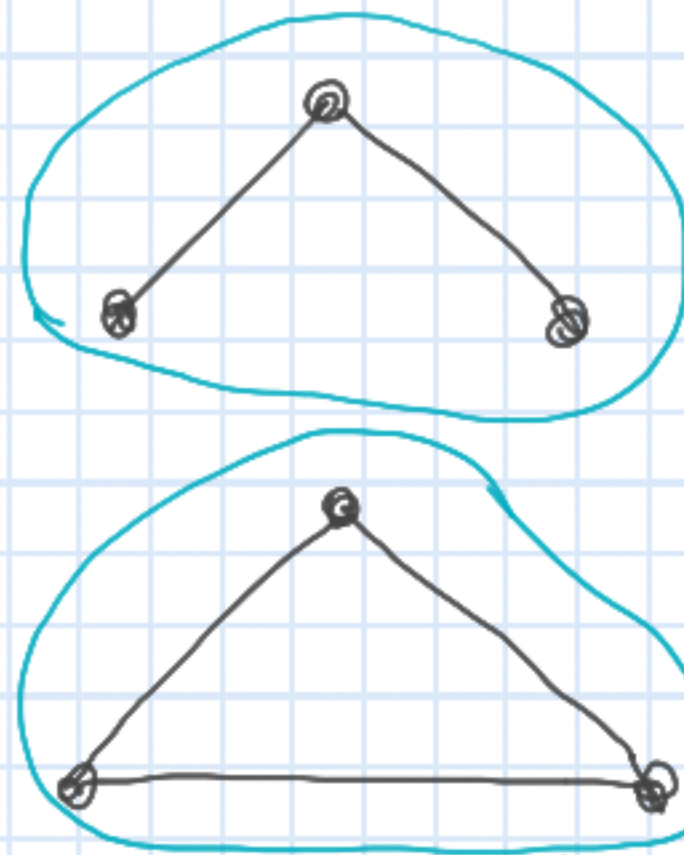
For Fun! 😊

Connectivity in undirected graphs

$G = (V, E)$ is a connected graph if any two vertices can be connected by a path in G .



connected

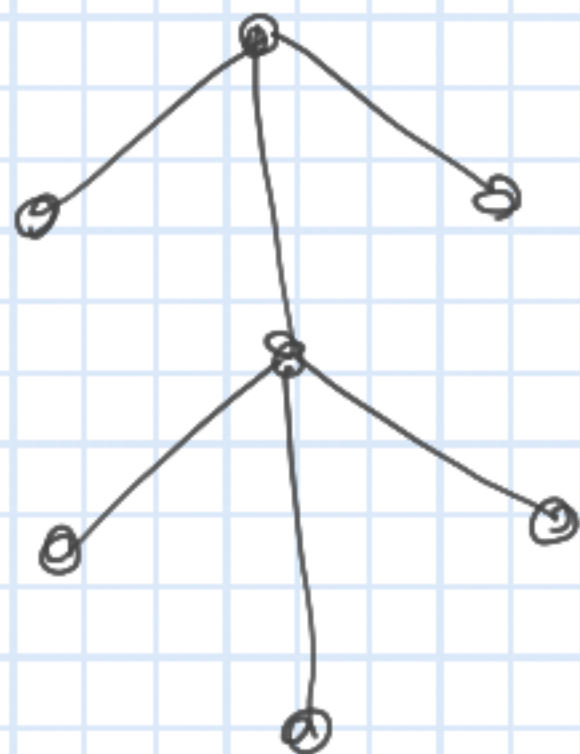


disconnected

connected components

Trees

undirected connected graph without cycles



$$|V| = |E| + 1$$

Representations of graphs

- adjacency lists
- adjacency matrix

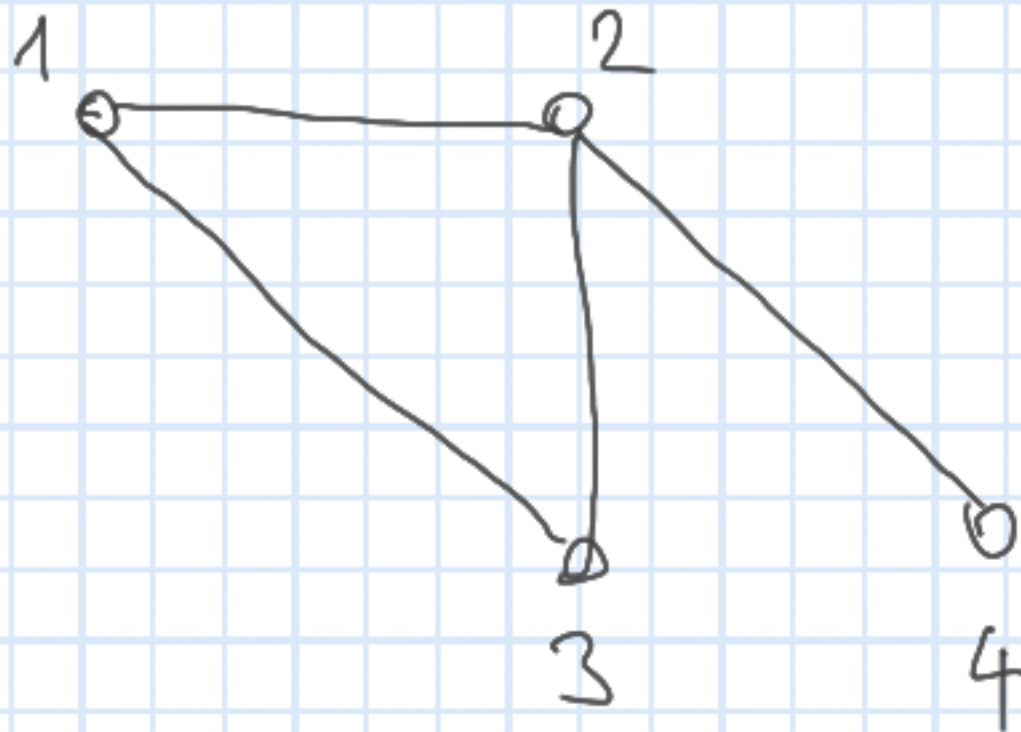
For the sake of simplicity we will assume that $V = \{1, 2, \dots, n\}$

Adjacency matrix

Here we represent a graph by a $|V| \times |V|$ dimensional matrix with entries 0 and 1 such that we have 1 in the cross of the i^{th} row and j^{th} column if i and j are connected by an edge and 0 otherwise

Esp. we have zeros in the main diagonal

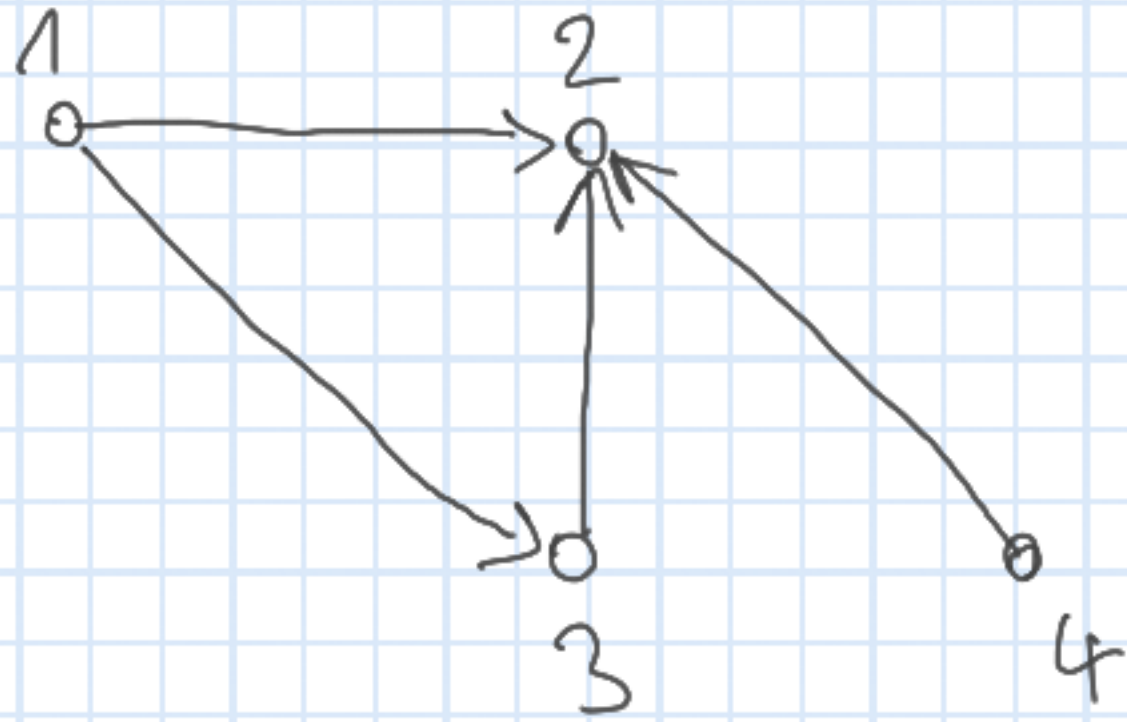
Example



undirected case

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

symmetric



directed graph

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

at the cross of the
 i^{th} row and j^{th} column
we have 1 if an
edge points to j from i
(non symmetric)

